

CRT - Answers are bolded

1. Use a decision structure to write an appropriate statement for each of the following:

a) Display Great job! when grade is 90 or higher.

```
if (grade >= 90) {  
    System.out.println("Great Job!");  
}
```

b) Display Error when number is less than 20 or greater than 50.

```
if (number >= 50 && number <= 20) {  
    System.out.println("Error");  
}
```

c) Add 2 to the value of y when y is less than 100.

```
if (y <= 100) {  
    y = y + 2;  
}
```

2. Assume num1 and num2 contain integer values.

Write an if-else if statement that displays one of the following messages as appropriate:

The first number is larger.

The second number is larger.

Numbers are equal.

```
if (num1 > num2) {  
    System.out.println("The first number is larger.");  
} else if (num2 > num1) {  
    System.out.println("The second number is larger.");  
} else {  
    System.out.println("Numbers are equal.");  
}
```

3.

a) Which is the appropriate word, odd or even for the blanks below?

```
if (num % 2 == 0) {  
    System.out.println("even number");  
} else {  
    System.out.println("odd number");  
}
```

b) Rewrite the if-else as a switch statement.

```
switch (num % 2) {  
    case 0:  
        System.out.println("even");  
        break;  
    default:  
        System.out.println("odd");  
        break;  
}
```

4. Write statements that use Math.random() to generate random numbers for each of the following situations:

a) Generate a random integer between 1 and 50.

```
System.out.println("Random number : " + (int)((50 - 1 + 1) * Math.random() + 1));
```

b) Generate a random integer between 20 and 100.

```
System.out.println("Random number : " + (int)((100 - 20 + 1) * Math.random() + 20));
```

c) Generate a random double between 10 and 20, inclusive.

```
System.out.println("Random number : " + (double)((20 - 10 + 1) * Math.random() + 10));
```

5. Identify the logic errors in the statements below, which should display a single appropriate message for any value of age:

```
if (age < 18) {  
    System.out.println("child");  
} else if (age >= 18 && age <= 65) {  
    System.out.println("adult");  
} else if (age > 65) {  
    System.out.println("senior");  
}
```

6. Given the following assignments, determine if each of the following expressions evaluates to

true or false:

size = 100 weight = 50 value = 75

a) size > 50 && weight == 50

True

b) value < 100 && !(weight == 50)

False

c) size >= 100 || value >= 100

True

d) weight < 50 || size > 50

True

e) !(value < 75)

True

f) !(size > 100 && weight > 50 && value > 75)

true

g) (value < 125 || weight < 76) && size == 100

True

8. Determine if each of the following are true or false. If false, explain why.

a) The condition of an if statement must be a Boolean expression.] **True**

b) A nested if statement and an if-else if statements are the same. **False because they both complete the same tasks but their flow /structure is different**

c) The expression in a switch statement must evaluate to a double. **False, switch statements normally need ints.**

d) Numbers generated by a computer program are actually pseudorandom. **True**

e) The (double) cast is needed to generate a random integer. **False because int and double are different things. Double creates a decimal, int creates an integer.**

f) A compound Boolean expression can contain more than two Boolean expressions. **True**

g) In a logical And expression, both operands must be true for the expression to evaluate to be true. **True**

h) In logical expressions, && is evaluated before !. **False, the ! is evaluated first.**

i) The pow() method in the Math class is used for exponentiation. **True**

j) The statement x = abs(-3); will return the value 3. **True**

l) A diamond shaped object represents a decision

in a flowchart.
True